23K-0062

GenAI - A1

Q:2

Forward Pass

$\therefore z_1 = W_1 X + b_1$

$$z_1 = \begin{bmatrix} .42 & .72 \\ 0 & .3 \\ .15 & .09 \end{bmatrix} \begin{bmatrix} .34 \\ .55 \end{bmatrix}$$

$$z_1 = \begin{bmatrix} .5388 \\ .165 \\ .1005 \end{bmatrix}$$

$\therefore y_1 = Relu(z_1)$

$$y_1 = \begin{bmatrix} .5388 \\ .165 \\ .1005 \end{bmatrix}$$

$\therefore z_2 = W_2 y_1$

$$z_2 = \begin{bmatrix} .44 & .03 & .55 \end{bmatrix} \begin{bmatrix} .5388 \\ .165 \\ .1005 \end{bmatrix}$$

$z_2 = [.297]$

$y_2 = [.297]$

$\therefore z_3 = W_3 y_2$

$$z_3 = \begin{bmatrix} .55 \\ .71 \\ .29 \end{bmatrix} [.297]$$

$$z_3 = y_3 = \begin{bmatrix} .163 \\ .21 \\ .086 \end{bmatrix}$$

$\therefore z_4 = W_4 y_3$

$$z_4 = \begin{bmatrix} .97 & .55 & .97 \\ .71 & .7 & .22 \end{bmatrix} \begin{bmatrix} .163 \\ .21 \\ .086 \end{bmatrix}$$

$$z_4 = y_4 = \begin{bmatrix} .357 \\ .281 \end{bmatrix}$$

$J(\theta) = -.34 \ln(.357) - .55 \ln(.28$

$J(\theta) = 1.048$

Backward Pass

$\therefore \nabla y_4 Div = y_4 - Actual$

$$= \begin{bmatrix} .357 \\ .281 \end{bmatrix} - \begin{bmatrix} .34 \\ .55 \end{bmatrix}$$

$$\nabla y_4 Div = \begin{bmatrix} .017 \\ -.269 \end{bmatrix}$$

$\therefore \nabla z_4 Div = \nabla z_4 y_4 \cdot \nabla y_4 Div$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} .017 \\ -.269 \end{bmatrix}$$

$$\nabla z_4 Div = \begin{bmatrix} .017 \\ -.269 \end{bmatrix}$$

CamScanner

$$\therefore \nabla W_4 Div = \nabla Z_4 Div \cdot Y_3^T$$

$$= \begin{bmatrix} .017 \\ -.269 \end{bmatrix} \begin{bmatrix} .163 & .21 & .086 \end{bmatrix}$$

$$\nabla W_4 Div = \begin{bmatrix} .002 & .003 & .001 \\ -.043 & -.056 & -.023 \end{bmatrix}$$

$$\therefore \nabla y_3 Div = W_4^T \cdot \nabla Z_4 Div$$

$$= \begin{bmatrix} .97 & .71 \\ .55 & .7 \\ .97 & .72 \end{bmatrix} \begin{bmatrix} .017 \\ -.269 \end{bmatrix}$$

$$\nabla y_3 Div = \begin{bmatrix} -.174 \\ -.178 \\ -.042 \end{bmatrix}$$

$$\therefore \nabla Z_3 Div = \nabla_{Z_3 y_3} \cdot \nabla y_3 Div$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -.174 \\ -.178 \\ -.042 \end{bmatrix}$$

$$\nabla Z_3 Div = \begin{bmatrix} -.174 \\ -.178 \\ -.042 \end{bmatrix}$$

$$\therefore \nabla W_3 Div = \nabla Z_3 Div \cdot Y_2^T$$

$$= \begin{bmatrix} -.174 \\ -.178 \\ -.042 \end{bmatrix} \begin{bmatrix} .297 \end{bmatrix} = \begin{bmatrix} -.051 \\ -.052 \\ -.012 \end{bmatrix}$$

$$\therefore \nabla y_2 Div = W_3^T \cdot \nabla Z_3 Div$$

$$= \begin{bmatrix} -.55 & .71 & .29 \end{bmatrix} \begin{bmatrix} -.174 \\ -.178 \\ -.042 \end{bmatrix} = \begin{bmatrix} -.234 \end{bmatrix}$$

$$\therefore \nabla Z_2 Div = \nabla_{Z_2 y_2} \cdot \nabla y_2 Div$$

$$= \begin{bmatrix} 1 \end{bmatrix} \begin{bmatrix} -.234 \end{bmatrix} = \begin{bmatrix} -.234 \end{bmatrix}$$

$$\therefore \nabla W_2 Div = \nabla Z_2 Div \cdot y_1^T$$

$$= \begin{bmatrix} -.234 \end{bmatrix} \begin{bmatrix} .5388 & .165 & .1005 \end{bmatrix}$$

$$\nabla W_2 Div = \begin{bmatrix} -.126 & -.038 & -.023 \end{bmatrix}$$

$$\therefore \nabla y_1 Div = W_2^T \cdot \nabla Z_2 Div$$

$$= \begin{bmatrix} .44 \\ .03 \\ .55 \end{bmatrix} \begin{bmatrix} -.234 \end{bmatrix} = \begin{bmatrix} -.102 \\ -.007 \\ -.128 \end{bmatrix}$$

$$\therefore \nabla Z_1 Div = \nabla_{Z_1 y_1} \cdot \nabla y_1 Div$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -.102 \\ -.007 \\ -.128 \end{bmatrix} = \begin{bmatrix} -.102 \\ -.007 \\ -.128 \end{bmatrix}$$

$$\therefore \nabla W_1 Div = \nabla Z_1 Div \cdot X^T$$

$$= \begin{bmatrix} -.102 \\ -.007 \\ -.128 \end{bmatrix} \begin{bmatrix} .34 & .55 \end{bmatrix}$$

$$\nabla W_1 \Delta v = \begin{bmatrix} -.034 & -.056 \\ -.002 & -.003 \\ -.043 & -.07 \end{bmatrix}$$

Updated Weights

$$W_4 = \begin{bmatrix} .969 & .598 & .969 \\ .731 & .728 & .231 \end{bmatrix}$$

$$W_3 = \begin{bmatrix} .575 \\ .736 \\ .296 \end{bmatrix}$$

$$W_2 = [.503 \quad .049 \quad .561]$$

$$W_1 = \begin{bmatrix} .437 & .748 \\ .001 & .301 \\ .171 & .125 \end{bmatrix}$$

Q:3

$$z_1 = Y_1 = \begin{bmatrix} .5388 \\ .165 \\ .1005 \end{bmatrix}$$

$$\therefore \mu' = W_\mu \cdot Y1$$

$$\mu' = [.5 \quad .5 \quad .5] \begin{bmatrix} .5388 \\ .165 \\ .1005 \end{bmatrix}$$

$$\mu = \mu' = [.402]$$

$$\therefore \sigma' = W_\sigma \cdot Y1$$

$$\sigma' = [.5 \quad .5 \quad .5] \begin{bmatrix} .5388 \\ .165 \\ .1005 \end{bmatrix}$$

$$\sigma = \sigma' = [.402]$$

$$\therefore z = \mu + \sigma \cdot \varepsilon$$

Taking $\varepsilon = 1$ here for demo, but actually $\varepsilon \sim N(0,1)$

$$z_2 = [.402] + [.402][1] = [.804]$$

$$\therefore z_3 = W_3 z_2$$

$$Y3 = z_3 = \begin{bmatrix} .55 \\ .71 \\ .294 \end{bmatrix} [.804] = \begin{bmatrix} .442 \\ .57 \\ .233 \end{bmatrix}$$

$$\therefore z_4 = W_4 Y_3$$

$$z_4 = Y_4 = \begin{bmatrix} .97 & .55 & .97 \\ .71 & .7 & .22 \end{bmatrix} \begin{bmatrix} .442 \\ .57 \\ .233 \end{bmatrix}$$

$$z_4 = Y_4 = \begin{bmatrix} .968 \\ .764 \end{bmatrix}$$

Recon Loss = $-.34 \ln(.968) - .55 \ln(.764)$

$= .159$

$KL = -\frac{1}{2}(1 + \log(.402^2) - .402^2 - .402^2)$

$KL = .572$

$\therefore J(\theta) = \text{Recon loss} + KL$

$J(\theta) = .159 + .572 = .731$

⇒ In Backward pass, we already take derivatives ($\nabla y_4 Div$, $\nabla z_4 Div$, $\nabla w_4 Div$, $\nabla z_3 Div$) will be calc in same way as done in AE. After it $\nabla w_3 Div$ will be calc using.

$\therefore \nabla w_3 Div = \nabla z_3 Div \cdot z_2^T$

Then, $\nabla z_2 Div = w_3^T \cdot \nabla z_3 Div$

$\therefore \nabla \mu' Div = 1 \times \nabla z_2 Div + M$

$\nabla \mu' Div = \nabla \mu' M \cdot \nabla \mu Div$

$= [1] \cdot \nabla \mu Div$

$\therefore \nabla \sigma Div = \varepsilon \cdot \nabla z_2 Div + (\sigma - 1/\sigma)$

$\nabla \sigma' Div = [1] \cdot \nabla \sigma Div$

$\therefore \nabla w_4 Div = \nabla \mu' Div \cdot y_1^T$

$\therefore \nabla w_\sigma Div = \nabla \sigma' Div \cdot y_1^T$

$\nabla y_1 Div = w_\mu^T \cdot \nabla_{\mu'} Div + w_\sigma^T \cdot \nabla_{\sigma'}$

$\nabla z_1 Div = \nabla_{z_1} y_1 \cdot \nabla y_1 Div$

$\nabla w_1 Div = \nabla z_1 Div \cdot x^T$

In the end $\nabla z_1 Div$ & $\nabla w_1 Div$ will also be calc similarly as done in backward pass in autoencoder.

We will calc all weights change matrices & update the weights.